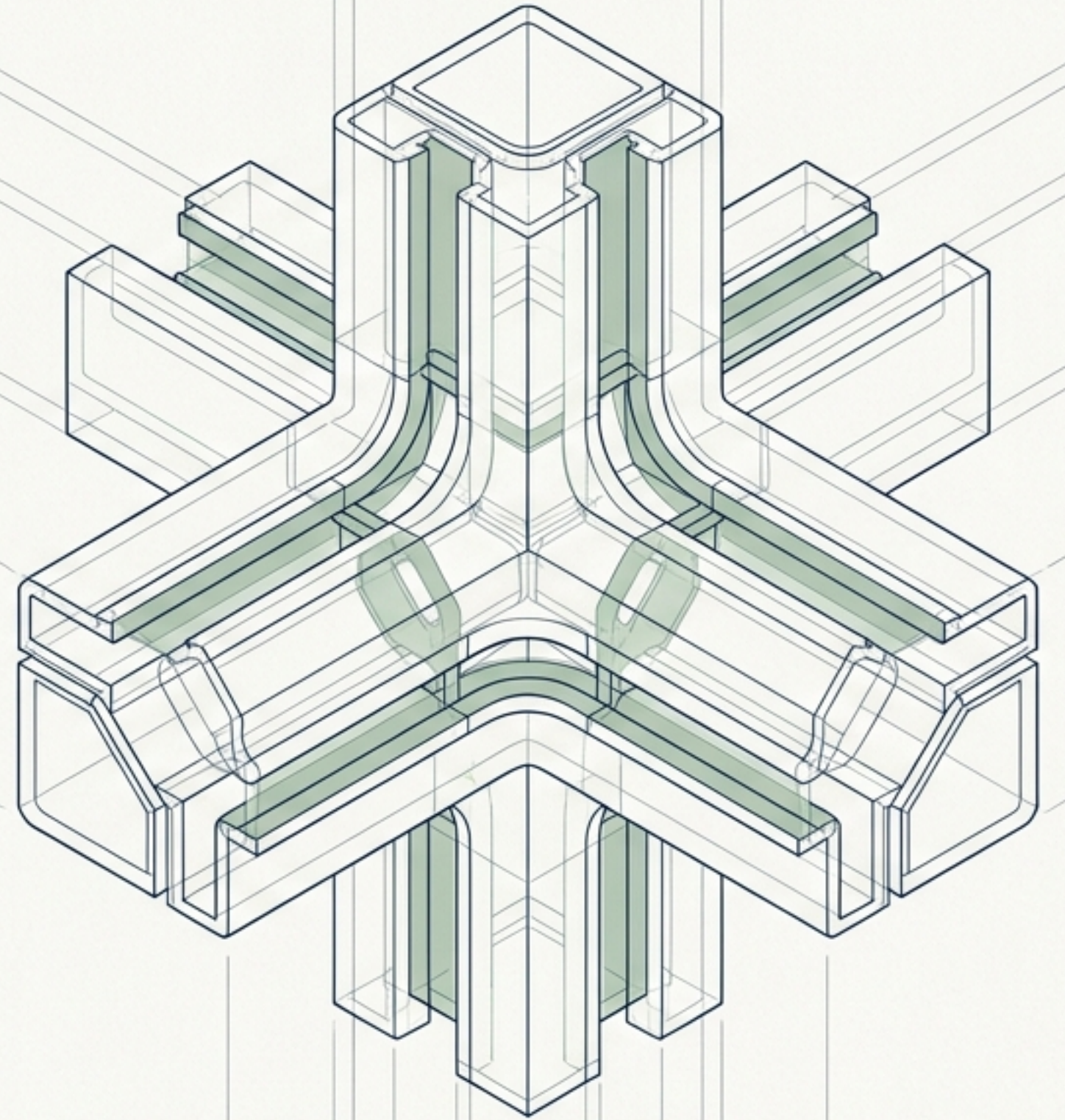


CCAR-P Exam Mastery: Tips, Tricks, and Traps

A tactical playbook and visual cheat sheet for navigating Anthropic's Professional architecture certification.

TARGET_SCORE: 800+ | STATUS: CONFIDENTIAL



THE EXAM LANDSCAPE & SOURCE OF TRUTH

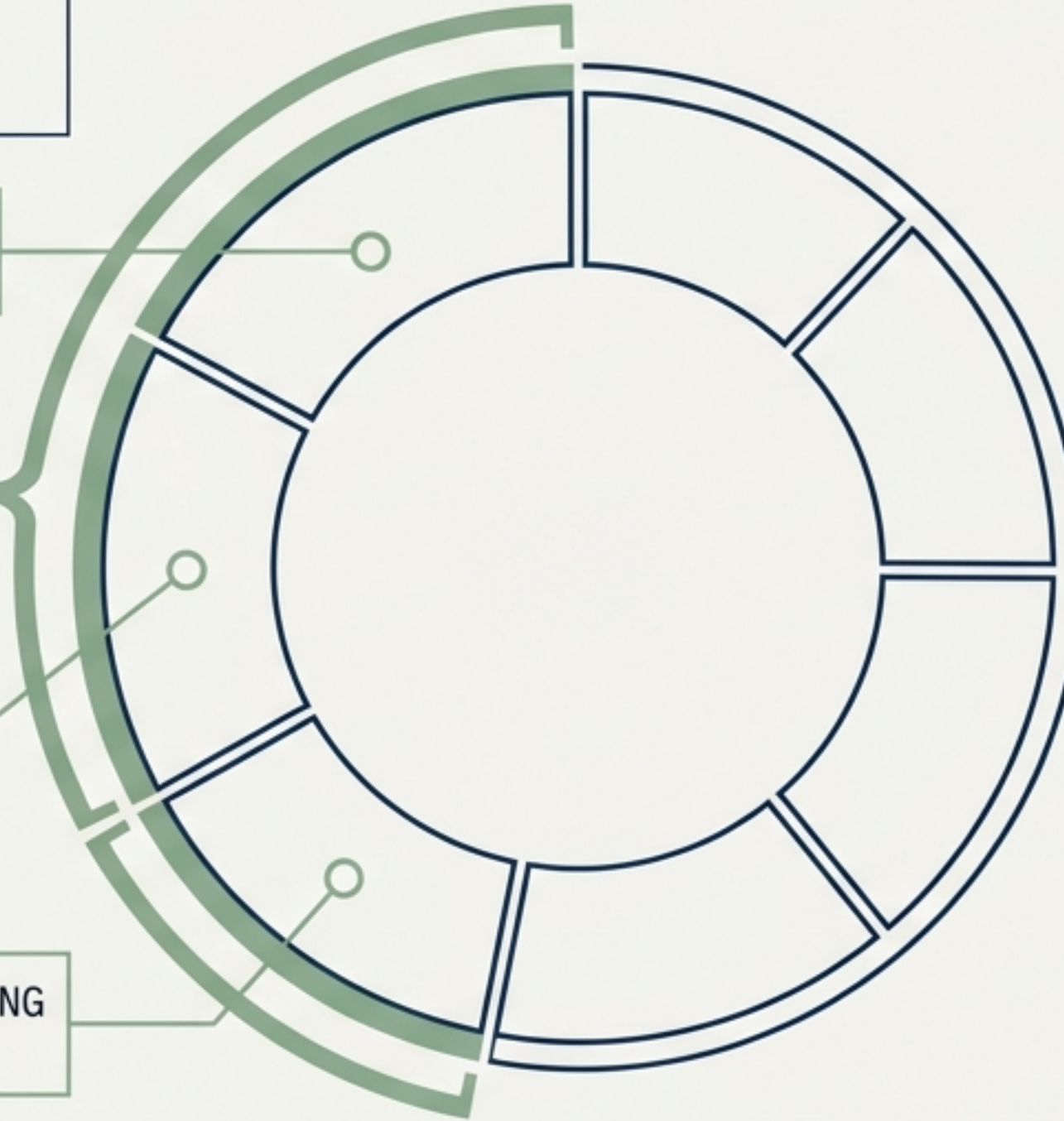
The Golden Rule: **The Official Exam Guide** always wins. If any other source disagrees, follow the **blueprint**.

DOMAIN 3:
INTEGRATION (19%)

THE 52% CORE

DOMAIN 1: SOLUTION DESIGN
& ARCHITECTURE (17%)

DOMAIN 4: EVALUATION, TESTING
& OPTIMIZATION (16%)



ITEMS	TIME
63 (MULTIPLE CHOICE/RESPONSE)	120 Minutes (~1.9 min/item)
CUT SCORE	PENALTY
720 / 1000	Unanswered = wrong (Always guess)

THE ARCHITECT'S DECODER: 6 MASTER PRINCIPLES



P1. Fix the Component

Does it change the thing that is actually broken?



P2. Least Privilege

Can capability be removed instead of guarded?



P3. Structural Optimization

Is there a design fix before using a blunt instrument?



P4. Proportionate Value

Is the solution sized to the risk?



P5. Governance by Design

Are controls built-in from the start?



P6. Evidence Over Intuition

Is the decision backed by metrics or logs?

TRAP: THE 8 DISTRACTOR CLASSES

Guard-Instead-of-Remove (Violates P2)

Adds a monitor to a capability that shouldn't exist.

Blunt-Instrument Optimization (Violates P3)

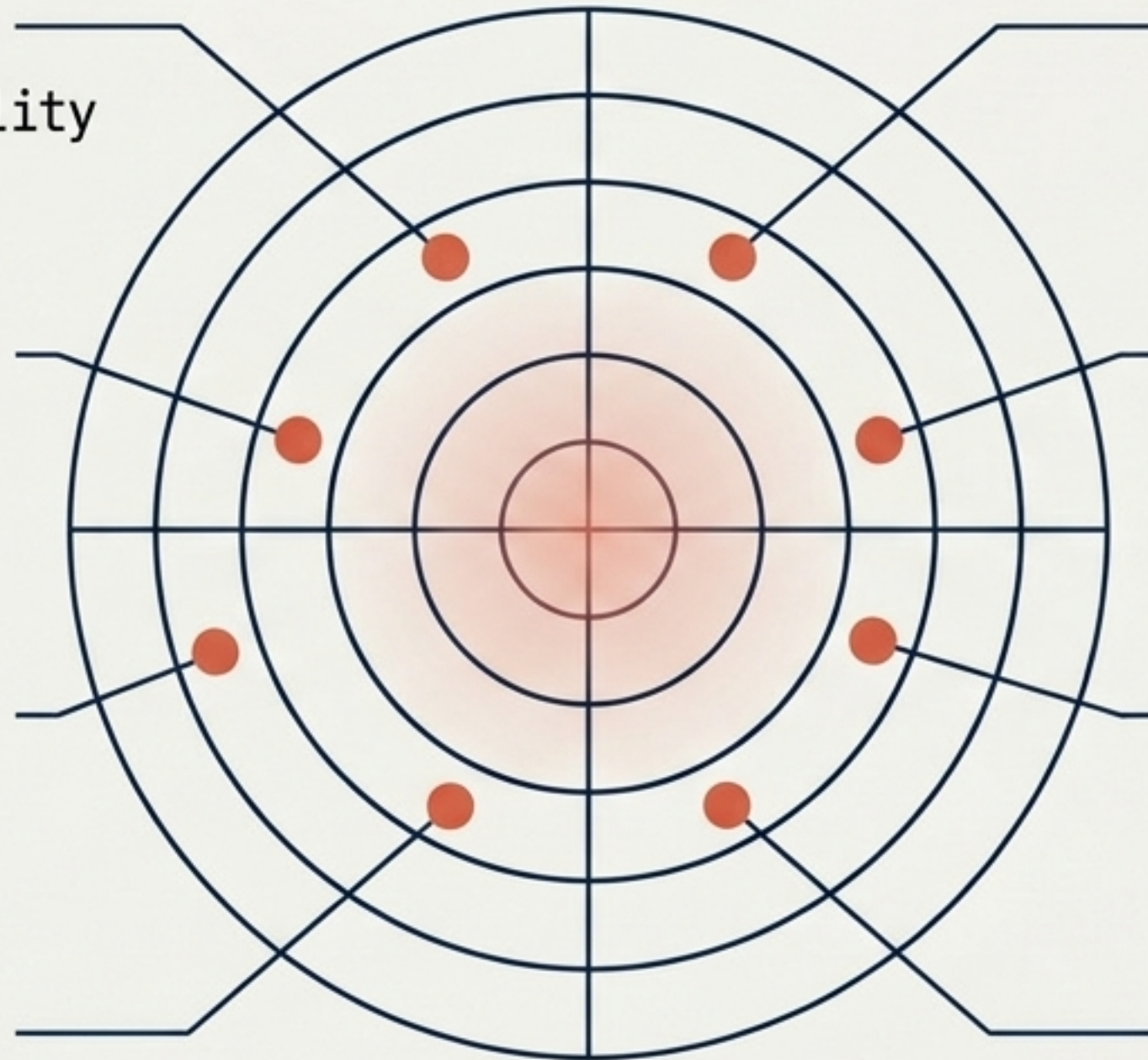
Blindly increasing max tokens or model size.

Wrong-Layer Diagnosis (Violates P1)

Tuning the prompt to fix a database timeout.

Detective-for-Preventive (Violates P5)

Alerting after a breach instead of blocking access.



Compliance-as-Afterthought (Violates P5)

Shipping first, adding PII handling later.

Vibes-Based Evaluation (Violates P6)

Relying on 'team feeling' instead of regression sets.

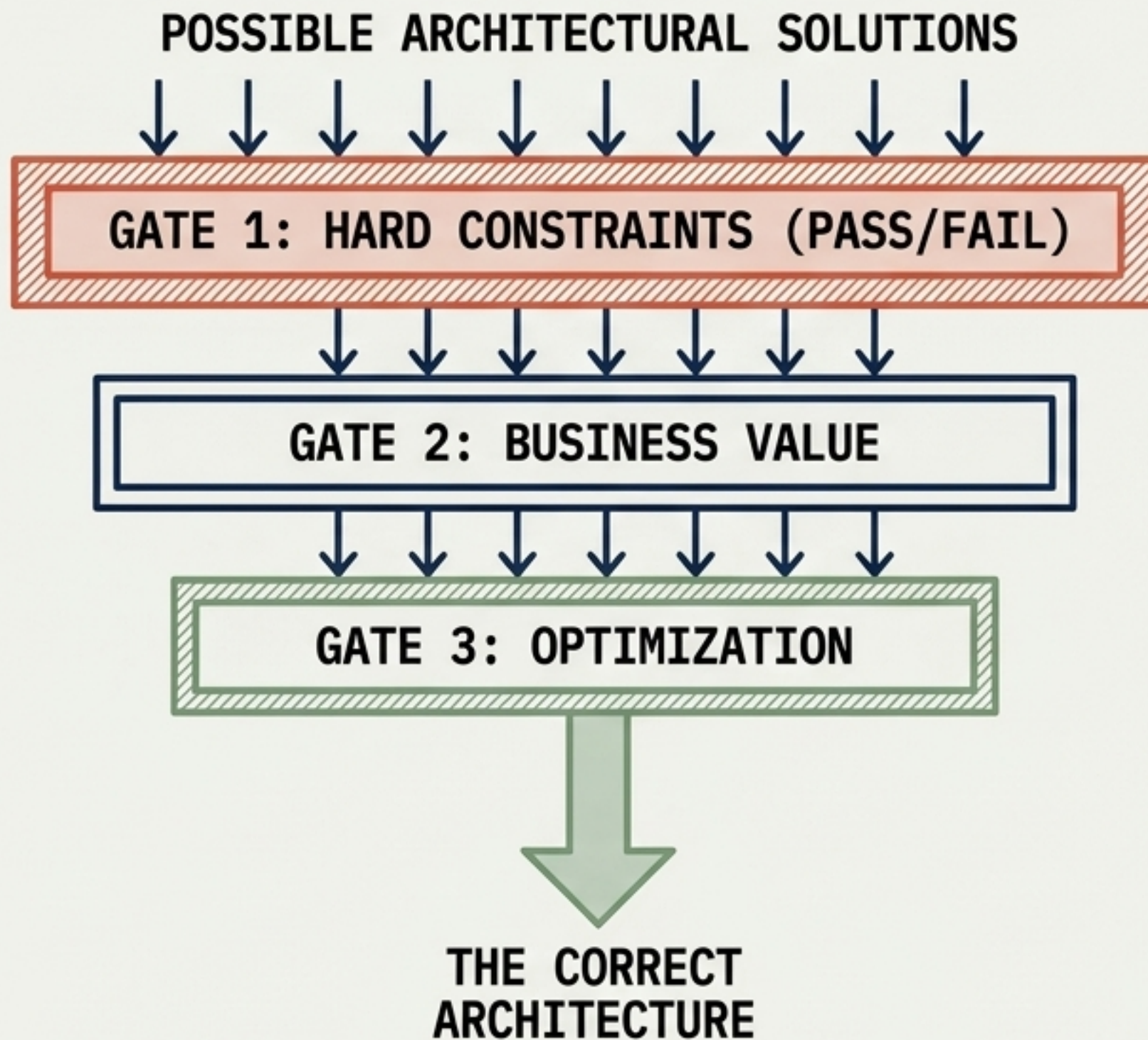
Capability Bloat (Violates P2)

Granting 'all tools' globally just in case.

Over-Engineering (Violates P4)

Deploying multi-agent swarms for simple routing.

EXECUTION STRATEGY: THE IRON TRIANGLE FILTER



Filter out any option violating **compliance, data boundaries, or SLA limits** (e.g., sub-50ms latency rules out LLMs).

Evaluate remaining by requested pillar: Efficiency, Productivity, or Transformation.

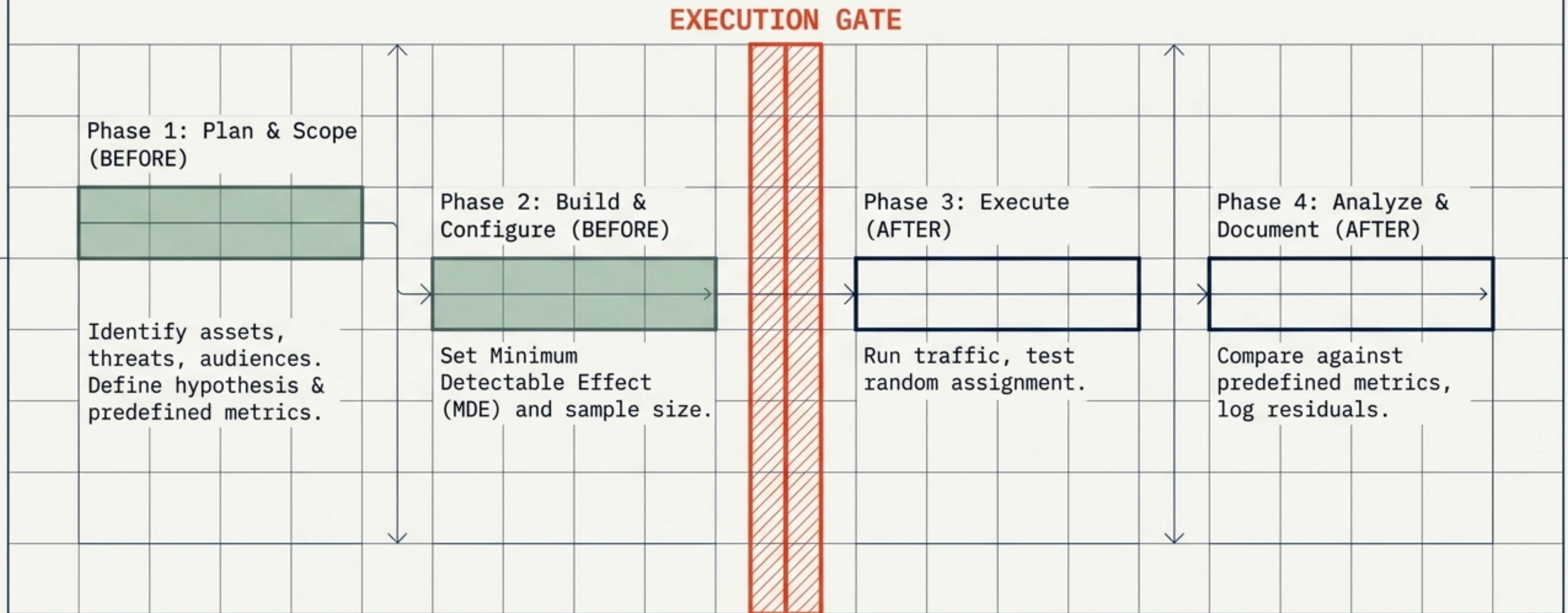
Pick the solution that structurally optimizes Cost/Latency without sacrificing Quality.

WATCH THE POLARITY:

Is the question asking for a valid configuration, or identifying a critical security gap?

TRAP: The BEFORE Timeline

When asked what to do BEFORE an event, eliminate options that require data from the event itself.



TRICK: The Claude Code Scope Hierarchy

Managed (Enforced)

IT/Admin pushed. Security baselines.
Cannot be overridden. Use for: Deny rules.

CLI (Session)

Command-line flags passed at execution.

Local (Personal Override)

`.claude/settings.local.json` (Git-ignored).
Applies to current repo for current user.

Project (Team Shared)

`.claude/settings.json` (Committed).
Use for: Shared MCP servers, team workflows.

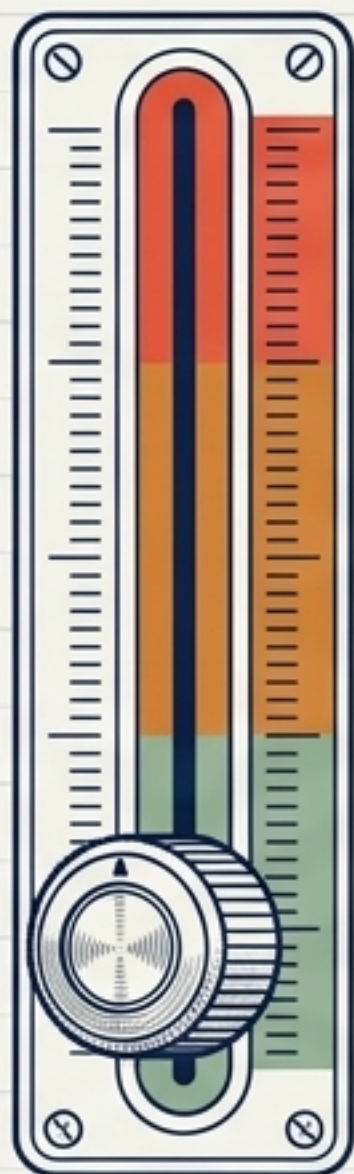
User (Personal)

`~/.claude/settings.json`.
Applies to all projects for one user.
Use for: Editor themes.

TRICK: Integration Routing Matrix

	Skill	MCP Server	Direct API
Nature	Procedural knowledge (how-to).	Live external data & tools.	Stateless, low-latency execution.
Contents	Templates, house style, checklists, examples.	DB queries, ticketing, CRM access, stateful tools.	Single HTTP call, pricing engines.
Rule of Thumb	Use when portability across clients is needed without external calls.	Use for live data requiring per-user authentication.	Use for simple, isolated functional endpoints.

TRAP: Capability Bloat & Least Privilege



Global / All Tools

Read + Send/Approve

Read + Write-to-Queue

Least privilege means removing capabilities, not monitoring them.

⚠ **Trap:** Guarding by the model ("tell prompt to ignore unneeded tools").

⚠ **Trap:** Logging destructive actions (Detective-for-preventive).

✓ **Tip:** Per-role allow-list enforced deterministically at orchestration layer

The "Draft" Rule: If a role drafts/proposes, grant read to knowledge and write to a review queue. Never grant send, approve, or delete.

TIP: The Optimization Toolkit

Tactic	Cost Impact	Latency Impact	Quality Impact	Verdict
Prompt Caching (Static Prefixes)	↓ DOWN	↓ DOWN (TTFT)	— EQUAL	Ultimate structural optimization.
Routing by Complexity	↓ DOWN	↓ DOWN	— EQUAL	Simple queries to small models.
Tool Search (Deferred Loading)	↓ DOWN	↓ DOWN	— EQUAL	Prevents context bloat.
Downsizing Everything	↓ DOWN	↓ DOWN	⚠ HIGH RISK	Blunt instrument; never do without eval.
Truncation / Summarization	↓ DOWN	↓ DOWN	⚠ HIGH RISK	Loses critical context.

TRAP: Vibes-Based Evaluation & Proxy Metrics

Vibes & Proxies (The Trap)

- **Metrics:** 89% First-Response Acceptance. Low line-count for PRs.
- **Risk:** Automation bias. Acceptance rate $\neq$ Correctness. Line count $\neq$ Risk.
- **Trap Answer:** "Ship it, acceptance proves readiness."

Outcomes & Evidence (The Fix)

- **Metrics:** Post-review override rate. Precision on fixed regression set.
- **Action:** Uncover the scope gap (what criteria is the AI missing that humans are checking?).
- **Rule:** Always evaluate against a fixed, version-attributed golden set before promotion.

TIP: Defense in Depth (Governance)

The control plane (identity/role) must be strictly separated from the data plane (message content).

Layer 1 (Outer): Audit: Actor attribution in tool-call logs.

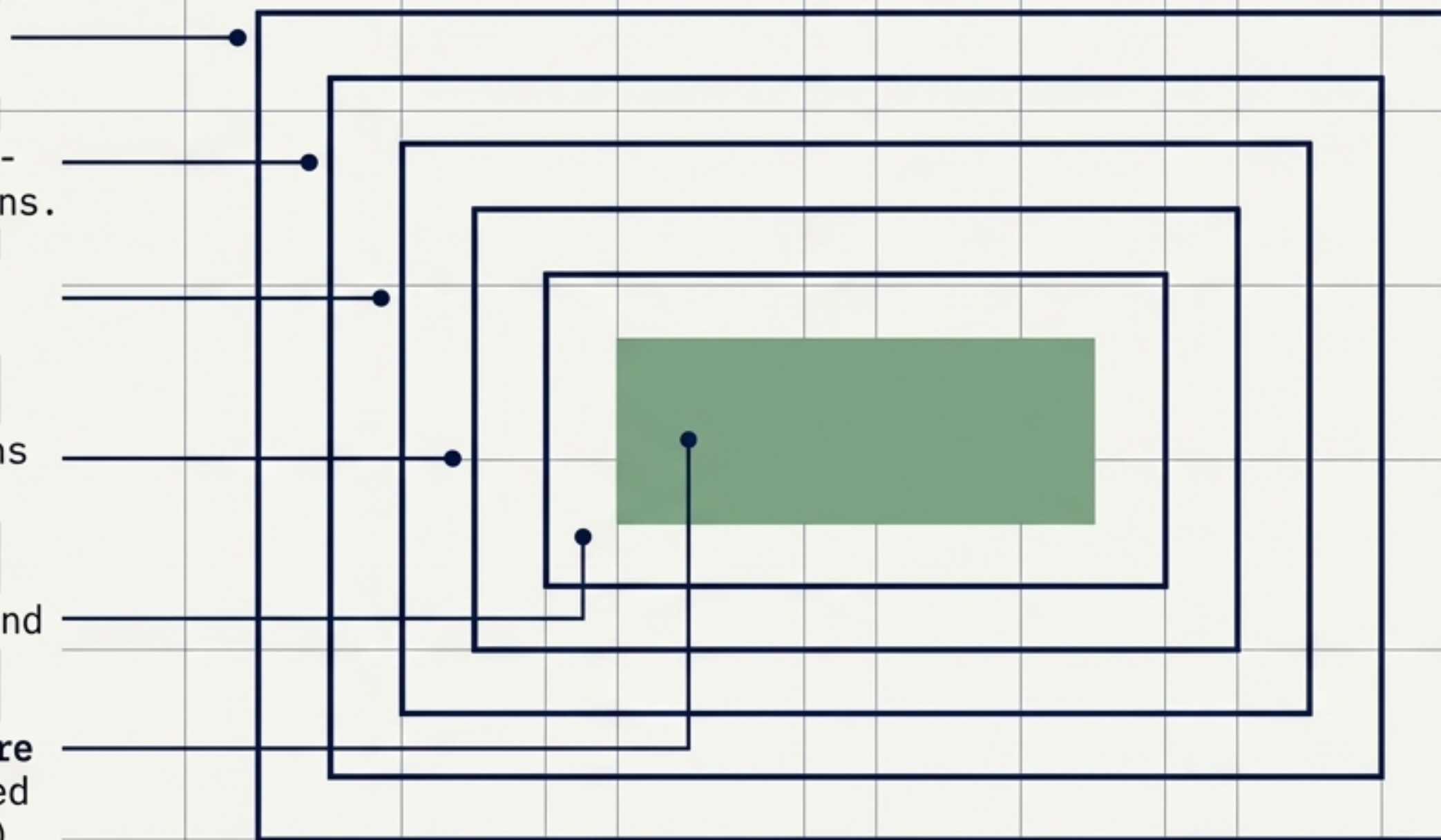
Layer 2: Action Gate: Human-in-the-loop (HITL) for irreversible actions.

Layer 3: Output Schema: Enforced JSON validation before action. (Deterministic)

Layer 4: Prompt/Model: Instructions and scoped constraints (Probabilistic).

Layer 5: Input Checks: Jailbreak and PII filters.

Layer 6 (Inner Core): Infrastructure Auth: Per-user OAuth, RBAC applied before retrieval. (Deterministic)



TRICK: Diagnosing LLM-Intrinsic Failure Modes

Would this failure still happen if you replaced the LLM with a normal deterministic function?

NO

YES

LLM-Intrinsic (Probabilistic)

- **Hallucination** (fluent but ungrounded).
- **Over-refusal** (triggering on safe keywords).
- **Over-refusal** (triggering on safe keywords).
- **Instruction drift** / 'Lost in the middle'.
- **Sycophancy & Indirect Prompt Injection**.

Generic Defect (Deterministic)

- Database connection timeouts.
- Expired TLS certificates.
- Load balancer routing to deprecated APIs.

TRICK: Context Strategy (RAG vs. Caching)

Corpus is millions of diverse documents.

One massive document (80K tokens) queried 14+ times.

60+ Internal Tools consuming budget.

RAG (Vector Index)

Retrieve only relevant chunks.
(Cost effective, scalable, risks missing cross-references).

Prompt Caching

Send full document as static prefix.
(Preserves 100% quality, drops cost on repeated reads, maintains strict data boundary).

Progressive Discovery (Tool Search)

Defer tool definitions until demanded.

Synthesis: The Production Iteration Loop

